

Weekly Internship Log

Week: 1

Student Name: 02240370

Company: DHI Drive

Supervisor: Lhendup Dorji

Day	Date	Start Time	End Time	Total Hours	Tasks/Activities/Accomplishments/Remarks
Friday	01/08/25	8:00 AM	5:00 PM	9 Hours	Researched on Nest JS as a backend framework
Monday	04/08/25	8:00 AM	5:00 PM	9 Hours	Created Nest JS app for backend project
Tuesday	05/08/25	8:30 AM	5:00 PM	8 Hours 30 Mins	Worked on Nest JS backend project
Wednesday	06/08/25	8:00 AM	5:00 PM	9 Hours	Worked on Nest JS backend project
Thursday	07/08/25	8:30 AM	5:00 PM	8 Hours 30 Mins	Attended a meeting SFL conference hall
Friday	08/08/25	8:00 AM	5:00 PM	9 Hours	Worked on Nest JS backend project

Total Weekly Hours: 53 Hours

Student's Weekly Summary & Reflection:

(What did you learn this week? What challenges did you face? What are your goals for next week?)

This week, I learned the fundamentals of Nest JS, including its modular structure, controllers, and services, and gained practical experience setting up a backend project and managing APIs. I faced some challenges understanding dependency injection and organizing the project structure efficiently, which required additional research and practice. For next week, my goal is to integrate a PostgreSQL database with the backend, develop API endpoints for data handling, and strengthen my understanding of backend authentication methods.

Supervisor's Comments/Feedback:

Supervisor's Signature: _____ **Date:** _____

Weekly Internship Log

Week: 2

Student Name: 02240370

Company: DHI Drive

Supervisor: Lhendup Dorji

Day	Date	Start Time	End Time	Total Hours	Tasks/Activities/Accomplishments/Remarks
Monday	11/08/25	8:00 AM	5:00 PM	9 Hours	Worked on Nest JS backend project
Tuesday	12/08/25	8:00 AM	5:00 PM	9 Hours	Researched awesome-nest-boilerplate in GitHub
Wednesday	13/08/25	8:00 AM	5:00 PM	9 Hours	Discussed with seniors about project structure and integration of boilerplate
Thursday	14/08/25	8:00 AM	5:00 PM	9 Hours	Merged existing NestJS project code into the new boilerplate structure
Friday	15/08/25	8:30 AM	5:00 PM	8 Hours 30 Mins	Reviewed integrated code with seniors and finalized backend feature improvements

Total Weekly Hours: 44.5 Hours

Student's Weekly Summary & Reflection:

(What did you learn this week? What challenges did you face? What are your goals for next week?)

This week, I learned to structure a NestJS project using a production-ready boilerplate, improving scalability and maintainability. With guidance from my seniors, I gained experience working with DTOs and ORMs like Prisma and TypeORM. The main challenge was adapting my old code to the new structure, but it helped strengthen my backend development skills. Next week, I aim to refine API endpoints and improve project documentation.

Supervisor's Comments/Feedback:

Supervisor's Signature: _____ **Date:** _____

Weekly Internship Log

Week: 3

Student Name: 02240370

Company: DHI Drive

Supervisor: Lhendup Dorji

Day	Date	Start Time	End Time	Total Hours	Tasks/Activities/Accomplishments/Remarks
Monday	18/08/25	8:00 AM	5:00 PM	9 Hours	Started new project under Senior Uzal; assisted colleague Pema Losel Maurer on education platform backend.
Tuesday	19/08/25	8:00 AM	5:00 PM	9 Hours	Developed and tested user registration and authentication endpoints using NestJS.
Wednesday	20/08/25	8:00 AM	5:00 PM	9 Hours	Began developing AI voicebot for user onboarding using Qwen3:32b model via Ollama.
Thursday	21/08/25	8:00 AM	5:00 PM	9 Hours	Implemented speech capture with MediaRecorder API and Whisper ASR for

					transcription; integrated ElevenLabs for TTS.
Friday	22/08/25	8:00 AM	5:00 PM	8 Hours 30 Mins	Consulted Senior Uzal for guidance and assisted at Super Fab Lab with sanding wooden boards for letter project.

Total Weekly Hours: 44.5 Hours

Student's Weekly Summary & Reflection:

(What did you learn this week? What challenges did you face? What are your goals for next week?)

This week, I learned to build and test backend authentication features in NestJS and integrate AI tools like Qwen3:32b, Whisper ASR, and ElevenLabs for a voicebot system. I also gained experience using browser APIs such as MediaRecorder for speech capture. The main challenge was synchronizing multiple AI services and ensuring smooth audio processing between frontend and backend. With guidance from Senior Uzal, I overcame these issues and improved my understanding of AI-driven backend integration. My goal for next week is to refine the voicebot's response flow and optimize API performance.

Supervisor's Comments/Feedback:

Supervisor's Signature: _____ **Date:** _____

Weekly Internship Log

Week: 4

Student Name: 02240370

Company: DHI Drive

Supervisor: Lhendup Dorji

Day	Date	Start Time	End Time	Total Hours	Tasks/Activities/Accomplishments/Remarks
Monday	25/08/25	8:00 AM	5:00 PM	9 Hours	Continued development of the AI Voicebot project with Pema Losel Maurer.
Tuesday	26/08/25	8:00 AM	5:00 PM	9 Hours	Faced and resolved GitHub merge conflicts caused by code synchronization issues.
Wednesday	27/08/25	8:00 AM	5:00 PM	9 Hours	Completed final development and debugging of the AI Voicebot system.
Thursday	28/08/25	8:00 AM	5:00 PM	9 Hours	Designed and implemented the Voicebot frontend using NextJS and Shadcn UI components.

Friday	29/08/25	8:30 AM	5:00 PM	8 Hours 30 Mins	Tested backend endpoints using Swagger UI and learned its documentation features.
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Total Weekly Hours: 44.5 Hours

Student's Weekly Summary & Reflection:

(What did you learn this week? What challenges did you face? What are your goals for next week?)

This week, I completed the AI Voicebot project and improved my skills in both backend and frontend development. I learned to resolve GitHub merge conflicts, design interfaces using NextJS and Shadcn UI, and test APIs through Swagger UI. The main challenge was handling merge errors caused by miscommunication, but working with my teammate helped me strengthen collaboration and version control skills. My goal for next week is to focus on enhancing API documentation and optimizing the overall project workflow.

Supervisor's Comments/Feedback:

Supervisor's Signature: _____ **Date:** _____

Weekly Internship Log

Week: 5

Student Name: 02240370

Company: DHI Drive

Supervisor: Lhendup Dorji

Day	Date	Start Time	End Time	Total Hours	Tasks/Activities/Accomplishments/Remarks
Monday	01/09/25	8:00 AM	5:00 PM	9 Hours	Assisted at Super Fab Lab (SFL) applying stickers on plastic sheets with embedded letters
Tuesday	02/09/25	8:00 AM	5:00 PM	8 Hours 30 Mins	Resumed work on the NestJS backend project; developed new routes and endpoints.
Wednesday	03/09/25	8:30 AM	5:00 PM	8 Hours 30 Mins	Tested and refined backend APIs using Swagger UI; reviewed feedback from Senior Lhendup and Senior Subham.
Thursday	04/09/25	8:30 AM	5:00 PM	8 Hours 30 Mins	Continued backend improvements and discussed optimization strategies with seniors.

Friday	05/09/25	8:00 AM	5:00 PM	9 Hours	Attended briefing on upcoming GMC Guest Management Portal project; assigned backend role and reviewed project tasks on Notion.
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Total Weekly Hours: 43.5 Hours

Student's Weekly Summary & Reflection:

(What did you learn this week? What challenges did you face? What are your goals for next week?)

This week, I strengthened my backend development skills by creating and testing APIs in NestJS and Swagger UI. I also learned about project management using Notion and gained experience improving existing codebases instead of starting from scratch. The main challenge was managing time between SFL tasks and development work, but it helped me improve my coordination and focus. My goal for next week is to begin development on the GMC Guest Management Portal and set up the backend structure efficiently.

Supervisor's Comments/Feedback:

Supervisor's Signature: _____ **Date:** _____

Weekly Internship Log

Week: 6

Student Name: 02240370

Company: DHI Drive

Supervisor: Lhendup Dorji

Day	Date	Start Time	End Time	Total Hours	Tasks/Activities/Accomplishments/Remarks
Monday	08/09/25	8:00 AM	5:00 PM	9 Hours	Began research on data centers in Bhutan with colleague Pema Losel Maurer under guidance from Senior Uzal.
Tuesday	09/09/25	8:00 AM	5:00 PM	9 Hours	Collected and reviewed information from BBS, Rest of the World, and other reliable sources..
Wednesday	10/09/25	8:00 AM	5:00 PM	9 Hours	Analyzed power consumption data from Bhutan Telecom annual reports.
Thursday	11/09/25	8:00 AM	5:00 PM	9 Hours	Studied DGPC's annual power production data and evaluated new hydropower projects.

Friday	12/09/25	8:00 AM	5:00 PM	9 Hours	Compiled research findings and began drafting the report advocating AI-powered data centers in Bhutan.
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Total Weekly Hours: 45 Hours

Student's Weekly Summary & Reflection:

(What did you learn this week? What challenges did you face? What are your goals for next week?)

This week, I focused on researching Bhutan's data centers and exploring the potential for AI-powered infrastructure. Working with my colleague, I learned to gather and analyze real-world data from verified sources such as BBS, DGPC, and Bhutan Telecom. I also improved my technical writing and analytical skills while studying power consumption and energy sustainability. The main challenge was interpreting large datasets accurately, but it helped me develop a stronger understanding of data-driven research. My goal for next week is to finalize the research report and prepare supporting visuals for presentation.

Supervisor's Comments/Feedback:

Supervisor's Signature: _____ **Date:** _____

Weekly Internship Log

Week: 7

Student Name: 02240370

Company: DHI Drive

Supervisor: Lhendup Dorji

Day	Date	Start Time	End Time	Total Hours	Tasks/Activities/Accomplishments/Remarks
Monday	15/09/25	8:00 AM	5:00 PM	9 Hours	Resumed data center research after midterm exams; continued data collection and analysis with colleague Pema Losel Maurer.
Tuesday	16/09/25	8:00 AM	5:00 PM	9 Hours	Created analytical visuals such as graphs and charts using real-world data.
Wednesday	17/09/25	8:00 AM	5:00 PM	9 Hours	Finalized report structure, added references, and ensured proper labeling of all data visualizations.
Thursday	18/09/25	8:00 AM	5:00 PM	9 Hours	Completed and submitted the final research report on AI-

					powered data centers in Bhutan.
Friday	19/09/25	8:00 AM	5:00 PM	9 Hours	Began work on Guest Management Portal; resolved GitLab cloning issue by learning SSH-based repository access.

Total Weekly Hours: 45 Hours

Student's Weekly Summary & Reflection:

(What did you learn this week? What challenges did you face? What are your goals for next week?)

This week, I focused on completing and refining the data center research report. I improved my skills in technical writing, data visualization, and organizing information with proper citations and formatting. After submitting the report, I transitioned to the Guest Management Portal project, where I learned how to clone GitLab repositories using SSH, strengthening my understanding of secure version control practices. The main challenge was adjusting back to work after midterm exams and resolving the Git cloning issue, but overcoming it helped build my confidence. My goal for next week is to fully set up the development environment and begin contributing to the backend for the Guest Management Portal.

Supervisor's Comments/Feedback:

Supervisor's Signature: _____ **Date:** _____

Weekly Internship Log

Week: 8

Student Name: 02240370

Company: DHI Drive

Supervisor: Lhendup Dorji

Day	Date	Start Time	End Time	Total Hours	Tasks/Activities/Accomplishments/Remar
Monday	22/09/25	8:00 AM	5:00 PM	9 Hours	Continued development on the Guest Management Portal project.
Tuesday	23/09/25	—	—	—	Public holiday (Blessed Rainy Day).
Wednesday	24/09/25	8:00 AM	5:00 PM	9 Hours	Began learning MongoDB through YouTube tutorials; received guidance from Senior Lhendup.
Thursday	25/09/25	8:00 AM	5:00 PM	9 Hours	Set up MongoDB Cloud database for the project; explored NoSQL concepts.
Friday	26/09/25	8:00 AM	5:00 PM	9 Hours	Worked on Notion tasks: modified schemas, cleaned up

					unused fields, and updated backend structure.
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Total Weekly Hours: 36 Hours

Student's Weekly Summary & Reflection:

(What did you learn this week? What challenges did you face? What are your goals for next week?)

This week, I focused on setting up and working with MongoDB for the Guest Management Portal project. Since I was new to MongoDB, I learned its basics through tutorials and guidance from Senior Lhendup, which helped me understand NoSQL databases and how they differ from relational systems. After configuring the cloud database, I worked on backend tasks such as updating schemas and cleaning up the codebase. The main challenge was learning a new database system from scratch, but it strengthened my ability to self-learn and adapt quickly. My goal for next week is to continue building out backend features and deepen my understanding of MongoDB's advanced capabilities.

Supervisor's Comments/Feedback:

Supervisor's Signature: _____ **Date:** _____

Weekly Internship Log

Week: 9

Student Name: 02240370

Company: DHI Drive

Supervisor: Lhendup Dorji

Day	Date	Start Time	End Time	Total Hours	Tasks/Activities/Accomplishments/Remarks
Monday	29/09/25	8:00 AM	5:00 PM	9 Hours	Began integrating Swagger UI into the Express.js backend for API documentation and testing.
Tuesday	30/09/25	8:00 AM	5:00 PM	9 Hours	Researched Swagger integration for Express; practiced through tutorials and mini-projects.
Wednesday	01/10/25	8:00 AM	5:00 PM	9 Hours	Completed Swagger setup and tested API endpoints; strengthened understanding through hands-on practice.
Thursday	02/10/25	8:00 AM	5:00 PM	9 Hours	Assigned new task on the official DHI website; reviewed GitLab

					repository and Figma design.
Friday	03/10/25	8:00 AM	5:00 PM	9 Hours	Implemented Figma changes: created carousel Hero section, added new GIFs/images, and added 10X Journey badge.

Total Weekly Hours: 45 Hours

Student's Weekly Summary & Reflection:

(What did you learn this week? What challenges did you face? What are your goals for next week?)

This week, I strengthened both my backend and frontend development skills. I learned how to integrate Swagger UI into an Express.js backend, which improved my understanding of API documentation and made testing endpoints much easier. Creating small practice projects helped reinforce these concepts and boosted my confidence. Later in the week, I transitioned to frontend tasks for the official DHI website, where I worked with GitLab and Figma to implement updated UI changes. The biggest challenge was adapting quickly between backend and frontend tasks, but it helped improve my versatility. My goal for next week is to continue refining the website changes and ensure that the updated design closely matches the Figma layout.

Supervisor's Comments/Feedback:

Supervisor's Signature: _____ **Date:** _____

Weekly Internship Log

Week: 10

Student Name: 02240370

Company: DHI Drive

Supervisor: Lhendup Dorji

Day	Date	Start Time	End Time	Total Hours	Tasks/Activities/Accomplishments/Remarks
Monday	06/10/25	8:00 AM	5:00 PM	9 Hours	Handed over DHI website project to colleague Yeshey; briefly resumed work on Guest Management Portal.
Tuesday	07/10/25	8:00 AM	5:00 PM	9 Hours	Assigned to new task: Research for RSTA/BCTA Recommendation Paper on number plate standards; began studying international number plate systems.
Wednesday	08/10/25	8:00 AM	5:00 PM	9 Hours	Continued research on design patterns, security features, readability, digital integration, and cost considerations.

Thursday	09/10/25	8:00 AM	5:00 PM	9 Hours	Assigned new task on the official DHI website; reviewed GitLab repository and Figma design.
Friday	10/10/25	8:00 AM	5:00 PM	9 Hours	Completed major sections of the Recommendation Paper; prepared for next week's presentation to the AD.

Total Weekly Hours: 45 Hours

Student's Weekly Summary & Reflection:

(What did you learn this week? What challenges did you face? What are your goals for next week?)

This week, I transitioned from development work to a research-focused assignment after handing over the DHI website to my colleague and briefly continuing the Guest Management Portal. Along with the other interns, I began preparing a Recommendation Paper for RSTA/BCTA, which required extensive research on international number plate standards, security features, readability requirements, and digital integration. We analyzed global systems, compared them with Bhutan's current approach, and created a modern prototype design in Canva, complete with updated color schemes, improved layouts, enhanced security features, and detailed justifications for every design choice. The main challenges were organizing large amounts of information and translating research findings into a practical, visually coherent design, but teamwork made the process smoother. By the end of the week, we started preparing our presentation for the AD next week, with goals to deliver the recommendations clearly, refine the paper based on feedback, and return to the Guest Management Portal afterward.

Supervisor's Comments/Feedback:

Supervisor's Signature: _____ **Date:** _____

Weekly Internship Log

Week: 11

Student Name: 02240370

Company: DHI Drive

Supervisor: Lhendup Dorji

Day	Date	Start Time	End Time	Total Hours	Tasks/Activities/Accomplishments/Remarks
Monday	13/10/25	8:00 AM	5:00 PM	9 Hours	Presented the Number Plate Recommendation Paper to the AD; assignment successfully concluded.
Tuesday	14/10/25	8:00 AM	5:00 PM	9 Hours	Continued finalizing the Guest Management Portal; sought clarification and feedback from Senior Lhendup.
Wednesday	15/10/25	8:00 AM	5:00 PM	9 Hours	Completed email logic for the Guest Management Portal; ensured integration with system workflow.
Thursday	16/10/25	8:00 AM	5:00 PM	9 Hours	Improved formatting of guest reports including layout restructuring, adding images, special

					occasions, and additional details.
Friday	17/10/25	8:00 AM	5:00 PM	9 Hours	Approached Senior Uzal to pursue AI-related tasks; assigned to work on AI agent using LangGraph; began learning fundamentals and researching workflow/state management concepts.

Total Weekly Hours: 45 Hours

Student's Weekly Summary & Reflection:

(What did you learn this week? What challenges did you face? What are your goals for next week?)

This week was productive and marked by both the completion of major tasks and the beginning of new learning opportunities. I successfully presented the Number Plate Recommendation Paper to the AD, officially concluding the assignment, and finalized the Guest Management Portal with guidance from Senior Lhendup, particularly in completing the email logic and improving the formatting of guest reports by adding images, special occasions, and other key details. A significant milestone was expressing my interest in AI-related work to Senior Uzal, after which I was assigned to begin learning LangGraph; I spent time understanding its workflow structure, state management, and integration with LLMs. The main challenges I faced this week included implementing clean and functional email logic, ensuring professional report formatting, and grasping LangGraph concepts from scratch. For next week, my goals are to continue building my understanding of LangGraph, begin experimenting with small AI agent prototypes, contribute meaningfully to the AI agent project, and support any required refinements to the Guest Management Portal.

Supervisor's Comments/Feedback:

Supervisor's Signature: _____ **Date:** _____

Weekly Internship Log

Week: 12

Student Name: 02240370

Company: DHI Drive

Supervisor: Lhendup Dorji

Day	Date	Start Time	End Time	Total Hours	Tasks/Activities/Accomplishments/Remarks
Monday	13/10/25	8:00 AM	5:00 PM	9 Hours	Continued studying LangGraph fundamentals; reviewed workflow structures and agent state patterns assigned by Senior Uzal.
Tuesday	14/10/25	8:00 AM	5:00 PM	9 Hours	Began drafting initial AI agent flow; experimented with node creation, transitions, and message routing in LangGraph.
Wednesday	15/10/25	8:00 AM	5:00 PM	9 Hours	Began a test project to learn and then designed a flow for the AI agent based on the test project.

Thursday	16/10/25	8:00 AM	5:00 PM	9 Hours	Built the router node for the AI agent to manage workflow routing between different modules.
Friday	17/10/25	8:00 AM	5:00 PM	9 Hours	Created the skeleton for the PDF parser; documented progress and prepared for integration with the AI agent workflow..

Total Weekly Hours: 45 Hours

Student's Weekly Summary & Reflection:

(What did you learn this week? What challenges did you face? What are your goals for next week?)

This week focused on advancing my work with the AI agent in LangGraph and laying the groundwork for additional functionalities. I continued exploring LangGraph fundamentals and workflow structures on Monday, followed by drafting the initial AI agent flow on Tuesday, experimenting with node creation, transitions, and message routing. By Wednesday, I implemented an early prototype and debugged state management issues, which helped solidify my understanding of agent workflows. On Thursday, I built the router node to manage workflow routing between different modules, and on Friday, I created the skeleton for the PDF parser, preparing it for future integration with the agent. The main challenges involved ensuring correct routing logic and designing the parser structure in a modular and extensible way. My goals for next week are to integrate the PDF parser with the AI agent, test end-to-end interactions, and refine the router node and workflow logic to improve stability and efficiency.

Supervisor's Comments/Feedback:

Supervisor's Signature: _____ **Date:** _____